Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/18/2012
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of the p53 Pathway in Spaceflight-Induced Tissue Degeneration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-135 ; Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our working hypothesis is that spaceflight factors (including radiation and microgravity) affect the growth and regenerative health of a broad range of mammalian tissues, including bone, muscle, and immune system, by causing indiscriminate slowing or arrest of normal cell proliferation. Specifically we hypothesize that there is a key regulatory cell-signaling pathway responsible for this space-induced arrest, namely the p53 signaling pathway, and that either radiation and/or microgravity can activate it. We are currently testing aspects of this hypothesis in ground studies, and propose to utilize mouse tissues from Bion M1 to test the p53 hypothesis with long-term spaceflight tissue samples. Specifically we propose to determine if the p53 pathway is activated during 30-day spaceflight in bone tissues by: 1) Biochemically measuring total and phospho- p53, MDM2, p21, PI3Kinase, and ubiquitinated p53 levels in bone tissue. 2) Immunohistochemically identifying specific cell types in bone with active p53 and DNA damage. 3) Correlating p53 pathway and related cell cycle status to bone loss measurements by histomorphometry, microCT and transmission R-ray microscopy. Because Bion M1 is planned to be the longest experimental spaceflight of mice to date, we expect to be able to measure clear spaceflight effects in tissue physiology and biochemistry, and to definitively be able to test the p53 hypothesis at hand. Because of our current NASA grant work in the p53 area of research, and because of experience in spaceflight experiments in Foton M2 and M3, we believe we have the expertise and experience to successfully analyze the samples we are requesting. The potential benefit of concluding these studies successfully is the identification of a common molecular mechanism to explain multiple spaceflight effects, which in turn could provide a valuable rationale for developing novel countermeasure targets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of the studies proposed is several-fold, but first and foremost they are a the test of an original and novel hypothesis that has the potential to be a common element of the already known tissue degenerative conditions observed in space, and an important predictor of yet unknown degenerative conditions that may emerge in long-term spaceflight, specially outside of LEO. There is already incontrovertible ground evidence that the p53 signaling pathway is key in the response to radiation, and there is also compelling evidence that p53 is required for tissue degenerative conditions caused by mechanical unloading. Although spaceflight evidence is limited, skin and muscle in rats flown in space also show elevated p53, strongly compelling us to follow this line of inquiry. The fact that p53 is a powerful tumor suppressor, and inducer of cell cycle arrest and programmed cell death, or apoptosis, should serve as clear indicator that in all likelihood, if spaceflight factors activate p53, this will have systemic effects on all the body processes involving cell proliferation. These processes are most notably the indispensable tissue regeneration that keep every organ from collapsing due to unrepaired cellular and tissue damage. In addition these studies are also highly significant because they attempt to pinpoint the cellular specificity of p53 activation in bone, and to determine if this is a organ-wide process or limited to certain cell types such as the bone marrow osteoprecursor somatic stem cell lineage. If p53 is proven to be significantly involved in spaceflight arrest of tissue regeneration and degenerative conditions, it will offer both a new pharmacological target for intervention by drugs such as pifithrin class inhibitors and as a biomarker of tissue damage in space. Finally the proposed studies will shed new light on the old question of bone loss in space, possibly extending our understanding of bone dynamics in adult animals, and by being the longest experimental rodent exposure to spaceflight to date. When these studies are concluded we expect that we will have gained new factual understanding of bone loss in space under conditions previously not attempted, and will have an important new perspective about the potential for p53 activation to be a serious problem for human tissue regeneration during long-term spaceflight, and in human health and disease related to disuse.
Task Progress:	During the period from July 2011-August 2012 we accomplished two main set of tasks, first and foremost preparing for the now scheduled May 2013 Bion M1 flight experiments, and second performing the analysis of mouse tissues provided to the Bion M1 investigators from the STS135 Mouse Biospecimen Sharing program. On the first task of preparing for Bion M1 operations in Moscow at the Institute for Biomedical Problems, we have now finalized and provided updated experimental management plans, reagent and supply lists, and shipping matrices. In addition we have started laboratory verification testing of the cell biology procedures with bone marrow we plan to conduct immediately following Bion M1 tissue recovery. Following the STS135 mouse BSP and despite no additional NASA funding being provided in 2011, we dissected and collected bone samples at KSC and managed to start analyzing the combined effects of microgravity and gamma irradiation on the p53 pathway in mouse metatarsals placed in organ culture immediately following spaceflight. Although the tissues we had requested (pelvis and femur) were also not provided, the metatarsals offered a good alternative model for a load bearing set of bones for analysis. Because of unfortunate errors in sample labeling and distribution, the n for the experimental tissues provided to BSP investigators was reduced, impacting our ability to perform all our planned analyses. However, due to the large number of metatarsal bones in the feet of each animal (10), we were able to conduct a slightly more limited but still statistically significant analysis of thickness, mineral content, and architectural differences in bones belonging to ground controls versus spaceflight, either un-irradiated or irradiated at 10 or 50 cGy. As a result we now know that cortical bone in metatarsals is sensitive to microgravity induced mineral loss, on the order of 6.4% per month in cortical thickness, with a significant p value of 0.0325. We also demineralized and sectioned additional metatarsal bones for immunohistochemical analysis of p53 pathway proteins, and recovered total RNA for gene expression studies, which are still ongoing. Once those studies are concluded we expect to determine if osteoclast maturation and recruitment to endosteal bone surfaces is subject to synergistic or additive effects of low-dose gamma irradiation and microgravity, and if activation of the p53 pathway by irradiation is modulated by microgravity exposure.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Almeida E. "Bone Loss in Microgravity is Associated with Activation of Specific Intracellular Signaling Pathways." ORS (Orthopaedic Research Society) 2012 Annual Meeting, San Francisco, CA, February 4-7, 2012. ORS (Orthopaedic Research Society) 2012 Annual Meeting, San Francisco, CA, February 4-7, 2012. ORS Annual Meeting Transactions, ISSN 0149-6433, 2012. , Feb-2012

Abstracts for Journals and Proceedings	Blaber E, Dvorochkin N, Yousuf R, Alwood J, Wang A, Globus R, Burns BP, Almeida EAC. "Mouse Bone Marrow Cells Exhibit Altered Expression of Differentiation Markers in Response to Spaceflight." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 77. , Nov-2011
Papers from Meeting Proceedings	Blaber E, Almeida EAC. "Stem Cell Tissue-Regenerative Health in Microgravity." ASGSB and ISGP Annual Meeting, San Jose, California, Nov 2 – 6, 2011. Journal of Gravitational Physiology, in press, as of June 2012. , Jun-2012